

SANGHYEON LEE

Email: shlee6825@kaist.ac.kr

Website: sites.google.com/view/sanghyeonlee/



EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

September 2021 – Present

Ph.D. in Artificial Intelligence

Advisor: Prof. Jaegul Choo

GPA: 3.0 / 4.3

Korea Advanced Institute of Science and Technology (KAIST)

September 2019 – February 2021

M.S. in Artificial Intelligence

Advisor: Prof. Jaegul Choo

GPA: 3.25 / 4.3

Seoul National University

February 2011 – August 2018

B.S. in Economics

GPA: 3.6 / 4.3

Including military service from 2013 to 2015

RESEARCH INTEREST

Human-in-the-loop AI, AI Agent for Visual Editing, Conditional Generative Model

PUBLICATIONS

(C: Conferences, W: Workshops, *: Equal contribution)

- [C5] **Sanghyeon Lee**, Minwoo Lee, Euijin Shin, Kangyeol Kim, Seunghwan Choi, and Jaegul Choo. OPRO: Orthogonal Panel-Relative Operators for Panel-Aware In-Context Image Generation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. Accepted (25.4% acceptance rate). [github]
- [C4] **Sanghyeon Lee**, Jooyeol Yun, and Jaegul Choo. Enabling Region-Specific Control via Lassos in Point-Based Colorization. *AAAI Conference on Artificial Intelligence (AAAI)*, 2025. Accepted (23.4% acceptance rate). [paper] [github]
- [C3] Jooyeol Yun*, **Sanghyeon Lee***, Minho Park*, and Jaegul Choo. iColoriT: Towards Propagating Local Hint to the Right Region in Interactive Colorization by Leveraging Vision Transformer. *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2023. Round 1 Accept (22% acceptance rate). [paper] [project page]
- [C2] Eungyeup Kim*, **Sanghyeon Lee***, Jeonghoon Park*, Somi Choi, Choonghyun Seo, and Jaegul Choo. Deep Edge-Aware Interactive Colorization against Color Bleeding Effects. *IEEE International Conference on Computer Vision (ICCV)*, 2021. **Oral Presentation** (3% acceptance rate). [paper] [project page]
- [C1] Youngwoo Cho*, Minwook Chang*, **Sanghyeon Lee**, Hyoungwoo Lee, Gerard Jounghyun Kim and Jaegul Choo. Efficient Adversarial Audio Synthesis via Progressive Upsampling. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020. (48.0% acceptance rate). [paper]

UNDER REVIEW

- [1] **Sanghyeon Lee***, Mina Hwang*, Soyoung Lee, Jaegul Choo, and YoungJoon Yoo. Iterative Coarse-to-Fine Refinement for Virtual Try-On with VLM-Guided Region Selection.

PROJECT

National Institute of Meteorological Research <i>Researcher</i>	South Korea 2024 – Present
Focus: Constructed dataset and implemented Retrieval-Augmented Generation (RAG) framework for measuring and enhancing forecasting accuracy.	
Electronics and Telecommunications Research Institute (ETRI) <i>Researcher</i>	South Korea 2022 – 2023
Focus: Developed a sketch-to-image colorization model based on point and reference inputs.	
Samsung Electronics <i>Researcher</i>	South Korea 2021
Focus: Developed a reference-based photo enhancement model.	

WORK EXPERIENCE

Korea Advanced Institute of Science and Technology <i>Teaching Assistant</i>	Seoul, South Korea September 2019 - Current
Naver Cloud <i>Research Intern</i>	Seongnam, South Korea July 2023 – December 2023
Focus: Develop the interactive matting model by utilizing the Segment-Anything (SAM)	
LG CNS <i>Research Assistant Intern</i>	Seoul, South Korea April 2019 – July 2019
Focus: Conducted literature reviews and provided research assistance on topics related to cloud gaming.	

SCHOLARSHIP

KAIST Support Scholarship, KAIST	September 2019 - Current
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AWARDS

1st Prize , Steinway & Sons Amateur Piano Competition	2022
Best Poster Award , KAIST AI Workshop	2022
Silver medalist , Korean Chemistry Olympiad	2006, 2007

PROGRAMMING SKILLS

Languages: Python, MATLAB
Technologies: PyTorch Ecosystem (PyTorch, PyTorch Lightning, Diffusers, vLLM, Accelerate, ...), AI Coding Tools (Cursor, Claude, ...)